

# Santa Fe synthetic control: measured results and a mean-field starting point

Reproducible analysis of the Cygnus runs

26 September 2026

## Abstract

This report describes two completed Santa Fe sweeps and a separate sample-size calibration. It records the exact simulation and analysis parameters, summarizes the observed finite-population outcomes, gives equations suitable for a mean-field approximation, and embeds every generated PNG in an indexed atlas. The model is synthetic and makes no LLM or provider calls. The information statistics are descriptive; they do not establish a causal controller effect.

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# 1 Run inventory and provenance

The original sweep covered 5 named beta pairs, 2 memory values, and 9 controller budgets: 90 cells, 100 independent episodes per cell, 9,000 episodes, and 279,000 canonical round rows. The expanded rho sweep used the same five beta pairs, budgets, episodes, and analysis settings at 11 memory values: 495 cells, 49,500 episodes, and 1,534,500 round rows. Each episode stores rounds 0–30. Both runs have a simulation seal and an information seal for every cell. The original run has 360 detectability rows; the expanded run has 1,980 (four conditioning variants per cell). Slurm finalizers 347 and 550 completed successfully. The calibration contains 2,800 subsampling repetitions and 14 summary rows; replacement finalizer 456 completed successfully after a concurrent recipe-write race was fixed. All values below are read from the saved Parquet or CSV results rather than inferred from plot colors.

| Run                     | Result root below /shared/home/cesar/work/results/studies/ | Final job |
|-------------------------|------------------------------------------------------------|-----------|
| Original beta/CMI       | santa_fe_beta_cmi/                                         | 347       |
| Sample-size calibration | santa_fe_sample_size_cmi/                                  | 456       |
| Rho phase sweep         | santa_fe_rho_phase/                                        | 550       |

The complete recipes are reproduced in Appendix A. Each simulation cell uses the same root seed, 20260925, but the episode seeds are generated from `SeedSequence([root seed, cell ID])`. Thus different parameter cells have independent random streams; cross-cell differences are not paired by a common episode realization.

## 2 Microscopic model and update order

There are  $N = 24$  agents and  $F = 10$  binary-signed facts. In each episode, seven facts have weight  $+1$  (truth) and three have weight  $-1$ , shuffled across fact identities. Each fact starts with three distinct random holders. An agent’s initial vote is drawn from its private evidence, without sampled messages. Truth is vote  $+1$ ; the controller advocates target vote  $-1$ .

At every round there are  $N$  asynchronous microscopic updates. Each slot picks one focal agent uniformly *with replacement*. The focal agent independently retains each active fact with probability  $\rho$ , samples  $q = 3$  messages without replacement from the *previous* board, and acquires any facts attached to those messages. If  $A_i$  is its active fact set after this update and  $w_f$  is a fact’s signed weight, its evidence signal is

$$e_i = \begin{cases} |A_i|^{-1} \sum_{f \in A_i} w_f, & |A_i| > 0, \\ 0, & |A_i| = 0, \end{cases}$$

$$s_i = \frac{1}{q} \sum_{m \in \text{sample}} v_m,$$

$$\Pr(v_i = +1 \mid e_i, s_i) = \text{sigmoid}(\beta_e e_i + \beta_s s_i), \quad \text{sigmoid}(z) = \frac{1}{1 + e^{-z}}.$$

The agent posts one public vote and one randomly selected active fact, if any. The  $N$  posts made during the round form the closed board. After all updates, the controller senses 12 of these posts without replacement (`sensing_fraction` = 0.5) and observes their target-vote share  $Y_t$ . It draws  $U_t \sim \text{Bernoulli}(\text{sigmoid}[8(0.5 - Y_t)])$ . If  $U_t = 1$  and the budget is positive, it appends  $b$  target-vote recommendations with *no facts* to the next board. No recommendation is posted at  $b = 0$ , even if the policy draw says act. The previous board remains fixed during the next round’s  $N$  updates; the new board replaces it only at round close.

The exact integer budget is  $b = \text{round}(24b_{\text{frac}})$ . The sweep uses  $b = 0, 3, 6, 9, 12, 15, 18, 21, 24$ . Board size after a round is  $24 + bU_t$ ; controller posts can change both the mean sampled vote and the fraction of sampled messages that carry facts. The controller never directly changes an agent’s private facts. Round zero records the initialized population and board; rounds 1–30 record the evolving state.

### 3 Parameter ledger

| Parameter               | Value and role                                                                                                                                    |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| $N, F, q, R$            | 24 agents; 10 facts; 3 previous-board messages per focal update; 30 rounds.                                                                       |
| Initial fact redundancy | 3 distinct holders per fact at initialization.                                                                                                    |
| Truth fact fraction     | 0.7, giving 7 positive and 3 negative facts in each episode.                                                                                      |
| $\rho$                  | Original: 0.75, 1.0. Expanded: 0.40, 0.46, 0.52, 0.58, 0.64, 0.70, 0.76, 0.82, 0.88, 0.94, 1.0. Per-fact retention probability on a focal update. |
| Budget fraction and $b$ | 0, 0.125, ..., 1.0; integer posts 0, 3, ..., 24.                                                                                                  |
| Controller              | Target $-1$ ; sense 12/24 closed-board messages; policy slope 8.0; target-share threshold 0.5.                                                    |
| Episodes and seed       | 100 per sweep cell; root seed 20260925; 4 simulation processes per task.                                                                          |
| Retention               | Canonical round trajectories saved; microscopic trajectories not saved; plots and reports saved.                                                  |

| Named beta pair      | $\beta_e$ | $\beta_s$ |
|----------------------|-----------|-----------|
| Competition low      | 0.75      | 1.00      |
| Competition baseline | 1.00      | 1.30      |
| Competition high     | 1.25      | 1.60      |
| Evidence dominated   | 2.00      | 0.50      |
| Social dominated     | 0.50      | 2.00      |

These five *pairs* are not a Cartesian grid of beta values. The fixed `model.rho`, beta, and budget entries in each YAML are base defaults; the listed sweep axes override them for physical cells.

#### 3.1 Information analysis and scheduler settings

Current and next target counts are exact integer counts; fact-coverage coordinates are discretized into six bins. The pooled actuation statistic is  $I(U_t; X_{t+1} | X_t)$  for  $t = 1, \dots, 29$ . The additional variants condition on binned mean agent fact coverage, binned population fact coverage, or both. The latter population coordinate occupies the shared estimator’s `phi` slot as a synthetic alias; it does not have HiddenBench `phi` semantics. Sensing MI is also computed for rounds 1–30. The recipe requests nine round statistics: sensing MI, target sensing MI, sensor/action MI, population, target, truth, mean-coverage-conditioned, population-coverage-conditioned, and jointly coverage-conditioned actuation CMI.

The shared estimator uses 200 whole-episode bootstrap resamples and 999 null draws per available pooled/per-round estimate, 95% confidence level, and analysis seed 20260925, with four analysis processes per task. The actuation null resamples actions from the recorded policy probability; its finite-draw  $p$ -value is  $(1 + \#\{T^* \geq T\}) / (1 + 999)$ . A separate legacy bootstrap (1000 draws, seed 43) and a separate legacy permutation null (1000 draws, seed 42) are both disabled. Histogram computation is disabled, though the configuration retains `save_null_histograms: true`. The configured sample-size section is disabled in both sweeps. Simulation and information arrays each request four CPUs and throttle at eight tasks; they request 8G for one hour and 12G for eight hours respectively. Finalizers request two CPUs, 32G, and two hours. These are execution settings, not scientific coordinates.

## 4 Measured outcomes

### 4.1 Budget response across beta regimes

The table shows final target-vote share averaged over the 11  $\rho$  values in the expanded sweep. Each entry is first averaged over 100 episodes within a cell, then averaged equally over the 11  $\rho$  cells. The last column is the mean of within- $\rho$  differences between  $b = 24$  and  $b = 0$ ; it is a descriptive finite-population contrast, without a cross-cell confidence interval.

| Beta regime          | $b = 0$ | $b = 24$ | Difference |
|----------------------|---------|----------|------------|
| Competition Low      | 0.373   | 0.498    | +0.124     |
| Competition Baseline | 0.312   | 0.489    | +0.177     |
| Competition High     | 0.252   | 0.494    | +0.242     |
| Evidence Dominated   | 0.287   | 0.356    | +0.069     |
| Social Dominated     | 0.332   | 0.614    | +0.282     |

The social-dominated pair has the largest mean budget contrast (+0.282); the evidence-dominated pair has the smallest (+0.069). The competition baseline is +0.177. The original two-rho sweep shows the same broad order, but it is a separate simulation with different cell IDs and random streams. Across all regimes and rho values, the expanded sweep’s mean target share is 0.312 at  $b = 0$  and 0.490 at  $b = 24$ ; the original sweep gives 0.306 and 0.489.

## 4.2 Memory, coverage, and target share

Across all five beta pairs and nine budgets, mean final individual fact coverage rises from 0.241 at  $\rho = 0.4$  to 0.996 at  $\rho = 1.0$ ; mean final population fact coverage rises from 0.441 to 1.000. In the same equal-cell average, final target share changes from 0.440 to 0.422. Thus memory strongly changes the epistemic state while the aggregate target-share response is much smaller and depends on beta regime and budget. These cell averages are not a thermodynamic order parameter or evidence of a sharp transition at  $N = 24$ . For example, at  $b = 24$  the social-dominated target share is 0.585 at  $\rho = 0.4$  and 0.614 at  $\rho = 1.0$ , while the evidence-dominated values are 0.355 and 0.337. The complete rho-by-budget maps are in the figure atlas.

## 4.3 Controller information and action support

At positive budget, all 440 expanded-sweep cells have both recorded action values, so pooled action CMI is defined. With 999 policy-null draws and an uncorrected  $p < 0.05$  threshold, the number of detectable cells varies with the conditioning set:

| Conditioning | Detectable/440 | Mean CMI minus null (bits) |
|--------------|----------------|----------------------------|
| Plain        | 318/440        | 0.035                      |
| Mean         | 294/440        | 0.032                      |
| Population   | 301/440        | 0.032                      |
| Both         | 274/440        | 0.028                      |

These are counts across a parameter sweep, not a multiple-testing-adjusted discovery count. The plain variant is detectable in 318/440 positive-budget cells; the joint coverage variant in 274/440. Corrected pooled CMI is often positive but can be slightly negative after null subtraction. The minimum null  $p$ -value is 0.001 at 999 draws. At  $b = 0$  the physical action never occurs, so actuation CMI, susceptibility, and its  $p$ -value are undefined. The effective-action field, not the policy’s latent  $U$  draw, is used. Conditional response  $\chi$  is the observed within-bin difference in next target share between action states, weighted by supported strata. Feedback confounding remains possible, so neither  $\chi$  nor CMI is a causal effect.

## 4.4 Sample-size calibration

The separate baseline-pair ( $\beta_e = 1.0, \beta_s = 1.3$ ),  $b = 6$  calibration simulated 400 reference episodes at each of  $\rho = 0.75$  and 1.0. At each rho, 200 repeated subsamples were drawn at  $n = 10, 20, 30, 50, 75, 100, 200$ . Each repetition requested 100 episode-bootstrap and 199 policy-null draws for plain target CMI. Detection means unadjusted null  $p < 0.05$ . Selected results:

| $n$ episodes | Detection at $\rho = .75$ | Detection at $\rho = 1$ | Note               |
|--------------|---------------------------|-------------------------|--------------------|
| 10           | 0.050                     | 0.040                   | 200 subsamples/rho |
| 50           | 0.195                     | 0.385                   | 200 subsamples/rho |
| 100          | 0.395                     | 0.790                   | 200 subsamples/rho |
| 200          | 0.880                     | 1.000                   | 200 subsamples/rho |

At  $n = 100$ , detection is 0.395 at rho 0.75 and 0.790 at rho 1.0 in this calibration, despite the main sweep using 999 rather than 199 null draws. The 200 subsamples repeatedly reuse one 400-episode reference population per

rho, so these fractions quantify detection in that reference, not universal power. They also do not calibrate the other nine rho values.

## 5 A mean-field approximation to try

Let  $p_t$  be the population truth-vote fraction,  $h_{f,t}$  the probability that a randomly chosen agent holds fact  $f$ , and  $m_t$  the mean vote among the  $N$  regular posts on the closed board. Tracking  $m_t$  separately from  $p_t$  matters because the  $N$  posts come from asynchronous draws with replacement. Given an effective controller action  $u_t \in \{0, 1\}$ , an approximate mean vote of the next board is

$$S_{t+1} \approx \frac{Nm_t - bu_t}{N + bu_t}, \quad \Pr(u_t = 1 \mid y_t) = \text{sigmoid}[8(0.5 - y_t)], \quad y_t \approx \frac{1 - m_t}{2}.$$

The last approximation neglects sensor sampling noise. A useful first closure can retain its binomial or hypergeometric distribution rather than replacing  $Y_t$  by its mean, because the sigmoid is nonlinear.

Suppose  $M_{f,t}$  of the  $B_t = N + bu_t$  previous-board messages carry fact  $f$ . The exact probability that a three-message sample contains fact  $f$  is

$$A_{f,t} = 1 - \frac{\binom{B_t - M_{f,t}}{q}}{\binom{B_t}{q}}, \quad q = 3.$$

An independence closure for one focal update is

$$h_{f,t}^{\text{new}} \approx \rho h_{f,t} + (1 - \rho) A_{f,t}.$$

This is the right place to model fact dilution by controller posts: they increase  $B_t$  but never  $M_{f,t}$ . It is also necessary to track at least the distribution of evidence balances, not only mean coverage, because  $\mathbb{E}[\text{sigmoid}(\beta_e e + \beta_s s)]$  generally differs from the sigmoid at the means. With a closure for that expectation, an approximate population equation over one round is

$$\frac{dp}{d\tau} = \mathbb{E}_{e,s}[\text{sigmoid}(\beta_e e + \beta_s s)] - p, \quad 0 \leq \tau \leq 1,$$

where one unit of  $\tau$  means  $N$  asynchronous slots. If the update target probability were constant during a round, this reduces to  $p_{t+1} \approx e^{-1}p_t + (1 - e^{-1})\mathbb{E}[\text{sigmoid}(\beta_e e + \beta_s s)]$ . That constant-field approximation is only a starting point: the code changes facts and votes during the round, even though it samples a fixed previous board. A stronger closure tracks vote/fact correlations and the distribution of regular posts as well as  $p_t$  and  $h_{f,t}$ . Fit or check such a closure against the saved time-resolved trajectories, not just the final heatmaps.

## 6 Interpretive limits

The finite design has  $N = 24$  and 100 episodes per cell. No thermodynamic limit, critical exponent, or sharp phase boundary was estimated. The rho-by-budget heatmaps are empirical response maps. Plot colors should be compared against the tables because each image can have its own color scale. Pooled CMI is calculated from pooled observations, not by averaging per-round estimates. Fact-coverage bins are six-level coarse variables, and the population coverage alias is specific to this synthetic adapter. The null test samples actions from the recorded stochastic policy; it does not randomize the controller itself. Statistical significance across hundreds of cells is unadjusted for multiplicity. The calibration reference is small relative to a universal power study. These limits matter when using the results to choose a mean-field closure or a future experiment.

## 7 Selected figures

The complete 634-image atlas is in the companion `santa_fe_results.pdf`; its index is `figure_manifest.csv`.

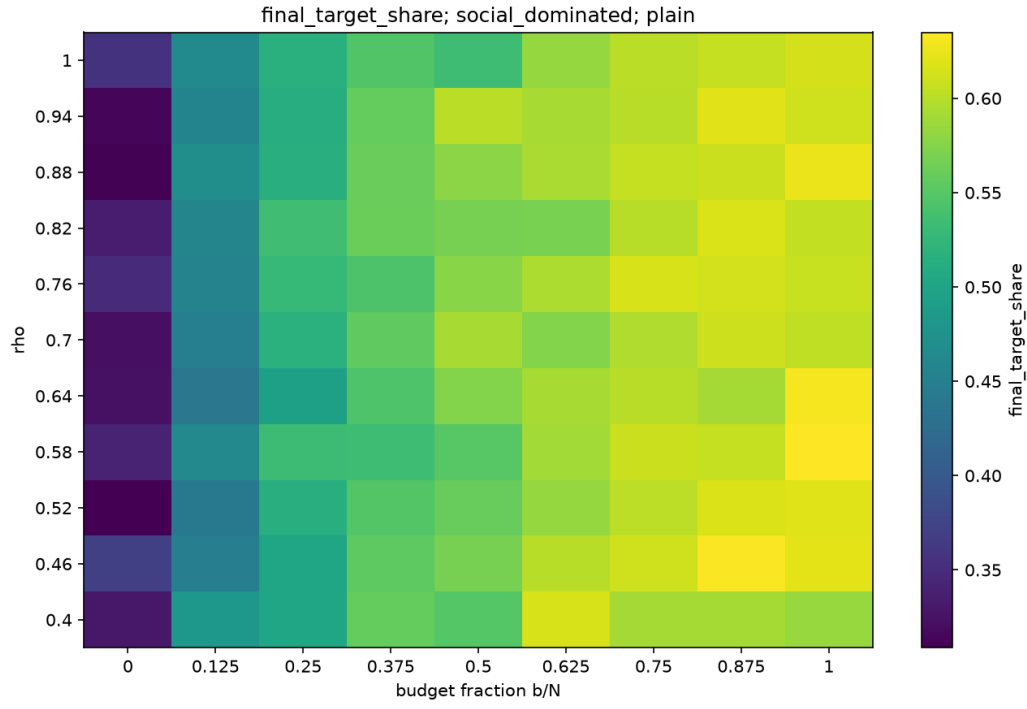


Figure 1: Final target share over rho and budget in the social-dominated beta regime.

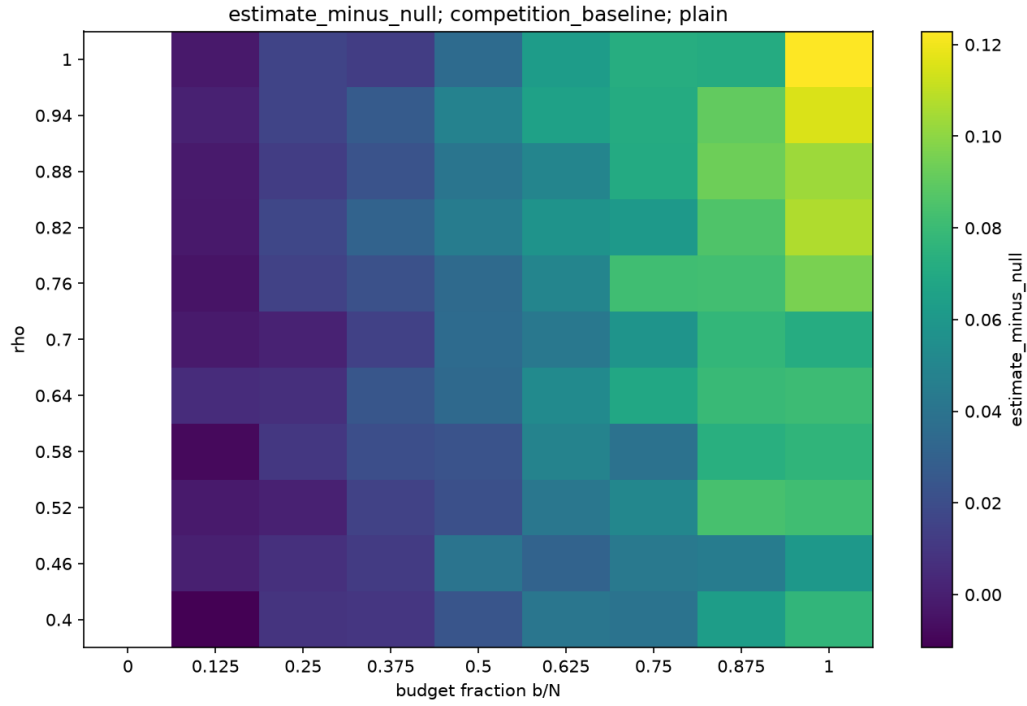


Figure 2: Plain target CMI minus policy-null mean in the baseline beta regime.

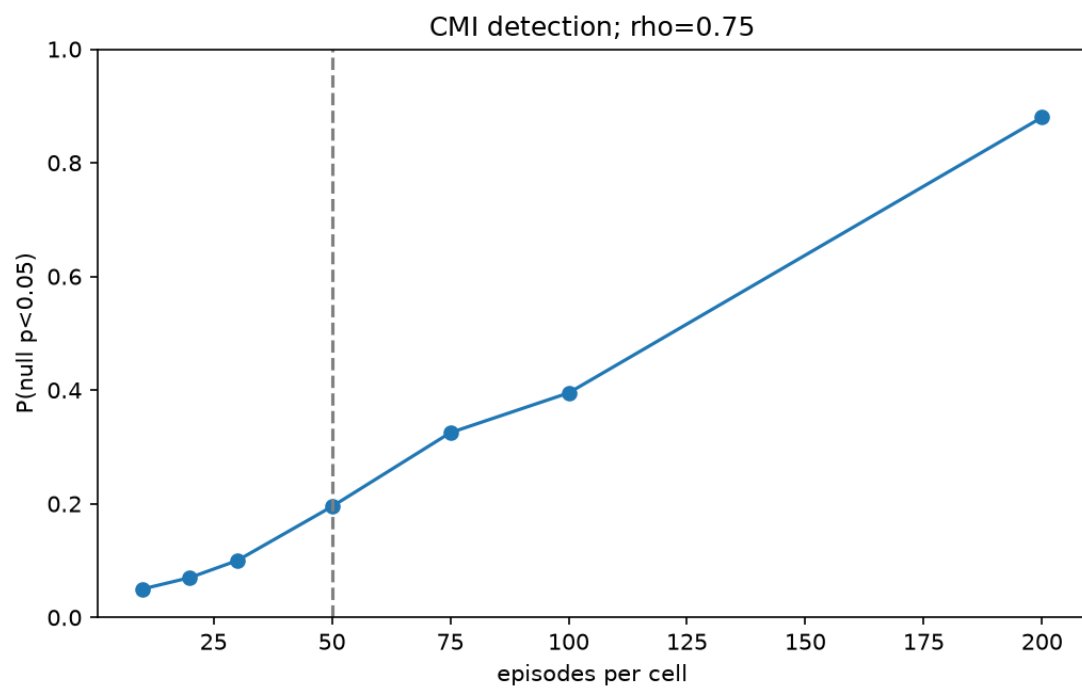


Figure 3: Sample-size detection curve at rho 0.75.

## A Exact input recipes

The full submitted YAML files follow. The analysis recipe and scheduler requests are retained here so fixed defaults, disabled switches, seeds, and resource requests remain visible alongside the scientific axes.

### A.1 beta\_cmi\_sweep.yaml

```
experiment:
  episodes: 100
  processes: 4
  seed: 20260925
output:
  results_dir: /shared/home/cesar/work/results/studies/santa_fe_beta_cmi
  save_round_trajectories: true
  save_micro_trajectories: false
  save_plots: true
  save_reports: true
model:
  N: 24
  F: 10
  q: 3
  rounds: 30
  initial_fact_redundancy: 3
  truth_fact_fraction: 0.7
  rho: 0.75
  beta_evidence: 2.0
  beta_social: 1.0
  sensing_fraction: 0.5
  controller_target: -1
  policy_beta: 8.0
  policy_threshold: 0.5
  budget_fraction: 0.125
sweep:
  beta_pairs:
    - name: competition_low
      beta_evidence: 0.75
      beta_social: 1.0
    - name: competition_baseline
      beta_evidence: 1.0
      beta_social: 1.3
    - name: competition_high
      beta_evidence: 1.25
      beta_social: 1.6
    - name: evidence_dominated
      beta_evidence: 2.0
      beta_social: 0.5
    - name: social_dominated
      beta_evidence: 0.5
      beta_social: 2.0
  rho:
    - 0.75
    - 1.0
  budget_fraction:
    - 0
    - 0.125
    - 0.25
    - 0.375
    - 0.5
    - 0.625
    - 0.75
    - 0.875
    - 1.0
analysis:
  bins: 6
  epistemic_coordinate: mean_coverage
  mutual_information:
    enabled: true
  permutation_null:
    enabled: false
    n_permutations: 1000
    seed: 42
  preserve_conditioning_state: true
bootstrap:
  enabled: false
  n_bootstrap: 1000
```

```

    unit: episode
    confidence_level: 0.95
    seed: 43
  llm_parallel:
    processes: 4
    bootstrap_resamples: 200
    null_permutations: 999
    confidence: 0.95
    seed: 20260925
    statistics:
      - round_sensing_mi
      - round_target_sensing_mi
      - round_sensor_action_mi
      - round_population_actuation_cmi
      - round_target_actuation_cmi
      - round_truth_actuation_cmi
      - round_kappa_target_actuation_cmi
      - round_phi_target_actuation_cmi
      - round_epistemic_target_actuation_cmi
  histograms:
    enabled: false
    save_null_histograms: true
  beta_cmi:
    enabled: true
  sample_size_study:
    enabled: false
    reference_episodes: 1000
    episode_counts:
      - 10
      - 20
      - 30
      - 50
      - 75
      - 100
    repetitions: 200
    n_bootstrap: 100
    n_permutations: 100
  slurm:
    simulation:
      cpus_per_task: 4
      memory: 8G
      time_limit: 01:00:00
      throttle: 8
    information:
      cpus_per_task: 4
      memory: 12G
      time_limit: 08:00:00
      throttle: 8
    finalize:
      cpus_per_task: 2
      memory: 32G
      time_limit: 02:00:00

```

## A.2 sample\_size\_cmi.yaml

```

analysis:
  beta_cmi:
    enabled: false
  bins: 6
  bootstrap:
    confidence_level: 0.95
    enabled: false
    n_bootstrap: 1000
    seed: 43
    unit: episode
  epistemic_coordinate: mean_coverage
  histograms:
    enabled: false
    save_null_histograms: true
  llm_parallel:
    bootstrap_resamples: 200
    confidence: 0.95
    null_permutations: 999
    processes: 4
    seed: 20260925
    statistics:

```

```

- round_sensing_mi
- round_target_sensing_mi
- round_sensor_action_mi
- round_population_actuation_cmi
- round_target_actuation_cmi
- round_truth_actuation_cmi
- round_kappa_target_actuation_cmi
- round_phi_target_actuation_cmi
- round_epistemic_target_actuation_cmi
mutual_information:
  enabled: true
permutation_null:
  enabled: false
  n_permutations: 1000
  preserve_conditioning_state: true
  seed: 42
experiment:
  episodes: 400
  processes: 4
  seed: 20260925
model:
  F: 10
  N: 24
  beta_evidence: 2.0
  beta_social: 1.0
  budget_fraction: 0.125
  controller_target: -1
  initial_fact_redundancy: 3
  policy_beta: 8.0
  policy_threshold: 0.5
  q: 3
  rho: 0.75
  rounds: 30
  sensing_fraction: 0.5
  truth_fact_fraction: 0.7
output:
  results_dir: /shared/home/cesar/work/results/studies/santa_fe_sample_size_cmi
  save_micro_trajectories: false
  save_plots: true
  save_reports: true
  save_round_trajectories: true
sample_size_study:
  enabled: true
  reference_episodes: 400
  episode_counts:
    - 10
    - 20
    - 30
    - 50
    - 75
    - 100
    - 200
  repetitions: 200
  n_bootstrap: 100
  n_permutations: 199
  statistic: round_target_actuation_cmi
slurm:
  finalize:
    cpus_per_task: 2
    memory: 32G
    time_limit: 02:00:00
  information:
    cpus_per_task: 4
    memory: 12G
    throttle: 8
    time_limit: 08:00:00
  simulation:
    cpus_per_task: 4
    memory: 8G
    throttle: 8
    time_limit: 01:00:00
sweep:
  beta_pairs:
    - name: competition_baseline
      beta_evidence: 1.0
      beta_social: 1.3
  rho:
    - 0.75

```

- 1.0
- budget\_fraction:
- 0.25

### A.3 rho\_phase\_sweep.yaml

```

experiment:
  episodes: 100
  processes: 4
  seed: 20260925
output:
  results_dir: /shared/home/cesar/work/results/studies/santa_fe_rho_phase
  save_round_trajectories: true
  save_micro_trajectories: false
  save_plots: true
  save_reports: true
model:
  N: 24
  F: 10
  q: 3
  rounds: 30
  initial_fact_redundancy: 3
  truth_fact_fraction: 0.7
  rho: 0.75
  beta_evidence: 2.0
  beta_social: 1.0
  sensing_fraction: 0.5
  controller_target: -1
  policy_beta: 8.0
  policy_threshold: 0.5
  budget_fraction: 0.125
sweep:
  beta_pairs:
    - name: competition_low
      beta_evidence: 0.75
      beta_social: 1.0
    - name: competition_baseline
      beta_evidence: 1.0
      beta_social: 1.3
    - name: competition_high
      beta_evidence: 1.25
      beta_social: 1.6
    - name: evidence_dominated
      beta_evidence: 2.0
      beta_social: 0.5
    - name: social_dominated
      beta_evidence: 0.5
      beta_social: 2.0
  rho:
    - 0.4
    - 0.46
    - 0.52
    - 0.58
    - 0.64
    - 0.7
    - 0.76
    - 0.82
    - 0.88
    - 0.94
    - 1.0
  budget_fraction:
    - 0
    - 0.125
    - 0.25
    - 0.375
    - 0.5
    - 0.625
    - 0.75
    - 0.875
    - 1.0
analysis:
  bins: 6
  epistemic_coordinate: mean_coverage
  mutual_information:
    enabled: true
  permutation_null:

```

```

    enabled: false
    n_permutations: 1000
    seed: 42
    preserve_conditioning_state: true
bootstrap:
    enabled: false
    n_bootstrap: 1000
    unit: episode
    confidence_level: 0.95
    seed: 43
llm_parallel:
    processes: 4
    bootstrap_resamples: 200
    null_permutations: 999
    confidence: 0.95
    seed: 20260925
    statistics:
        - round_sensing_mi
        - round_target_sensing_mi
        - round_sensor_action_mi
        - round_population_actuation_cmi
        - round_target_actuation_cmi
        - round_truth_actuation_cmi
        - round_kappa_target_actuation_cmi
        - round_phi_target_actuation_cmi
        - round_epistemic_target_actuation_cmi
histograms:
    enabled: false
    save_null_histograms: true
beta_cmi:
    enabled: true
sample_size_study:
    enabled: false
    reference_episodes: 1000
    episode_counts:
        - 10
        - 20
        - 30
        - 50
        - 75
        - 100
    repetitions: 200
    n_bootstrap: 100
    n_permutations: 100
slurm:
    simulation:
        cpus_per_task: 4
        memory: 8G
        time_limit: 01:00:00
        throttle: 8
    information:
        cpus_per_task: 4
        memory: 12G
        time_limit: 08:00:00
        throttle: 8
    finalize:
        cpus_per_task: 2
        memory: 32G
        time_limit: 02:00:00

```